Open-Source Software Audit

Comprehensive Technical Evaluation of MySQL 8.0

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Contents

Introduction and Project Scope

1. PART A1: THE ORIGIN STORY OF MYSQL

1.1 Etymology and Naming

1.2 The LAMP Stack Revolution

1.3 Historical Evolution & Corporate Journey

2. PART A2: LICENSING & OPEN-SOURCE PHILOSOPHY

2.1 The GNU General Public License (GPL v2)

2.2 The Commercial Track

3 Part A3: The Ethics of Open Source

4. PART B: LINUX FOOTPRINT & ARCHITECTURE

4.1 Binary and Executable Paths

4.2 Configuration and Data Storage

4.3 User, Group, and Security Context

4.4 The Update and Patch Model

5. Part C: Community Engagement

5.1 Forks and Ecosystem

6. Part D: Open Source vs. Proprietary Analysis

6.1 Overall Verdict

7. Part E: Project Execution

7.1 Script 1: System Identity

7.2 Script 2: Software Installation Audit

7.3 Script 3: Directory Mapping

7.4 Script 4: Storage Consumption

7.5 Script 5: Permissions Audit

8. Conclusion

Introduction and Project Scope

The rapid proliferation of Open-Source Software (OSS) has fundamentally transformed the landscape of modern computer science and software engineering. Unlike proprietary systems, OSS provides developers with the transparency to audit, modify, and redistribute code, fostering an environment of collaborative innovation.

1. PART A1: THE ORIGIN STORY OF MYSQL

The inception of MySQL traces back to the mid-1990s in Sweden. It was co-founded by Michael “Monty” Widenius, David Axmark, and Allan Larsson. The software was originally designed as a high-performance alternative to the proprietary mSQL database, which lacked the necessary speed for the massive datasets the founders were handling at their company, TcX.

1.1 Etymology and Naming

A common misconception is that “My” refers to the English possessive pronoun. In reality, the software is named after Michael Widenius's daughter, My Widenius. This tradition of naming forks after his children continued with the creation of MariaDB (named after his younger daughter, Maria).

1.2 The LAMP Stack Revolution

From its release in 1995, MySQL rapidly became the database of choice for the early web. It was the “M” in the famous LAMP Stack (Linux, Apache, MySQL, PHP/Python/Perl). This combination allowed developers to build dynamic websites using entirely free and open-source software, breaking the monopoly of expensive proprietary databases like Oracle or IBM DB2.

1.3 Historical Evolution & Corporate Journey

- **2008:** Sun Microsystems acquired MySQL AB for approximately $1 billion.
- **2010:** Oracle Corporation acquired Sun Microsystems, bringing MySQL under the control of a proprietary giant. This led to significant community debate regarding the future of the open-source version.

2. PART A2: LICENSING & OPEN SOURCE PHILOSOPHY

MySQL follows a Dual-License model, which is one of the most successful business strategies in the history of open-source software.

2.1 The GNU General Public License (GPL v2)

The primary version of MySQL is distributed under the GPL v2. This is a “Copyleft” license, which ensures that:

- Freedom to Run: Users can run the software for any purpose.
- Freedom to Study: The source code is freely available for audit.
- Reciprocity: If a developer modifies MySQL and distributes it, they *must* release their changes under the same GPL license. This prevents proprietary "locking" of community improvements.

2.2 The Commercial Track

Oracle also offers a commercial license. This allows companies to integrate MySQL into proprietary software without being forced to open-source their own code. This revenue helps fund the continuous development of the free community version.

3 Part A3: The Ethics of Open Source

The philosophy of Open Source is not merely a legal framework but an ethical stance on the democratization of knowledge. As an auditor, I have reflected on the following key ethical dimensions:

1. Should all software be open source? While the ideal of universal open source promotes maximum innovation, there is a pragmatic argument for proprietary code in high-stakes security sectors. However, for foundational infrastructure like databases (MySQL) and operating systems (Linux), open source is ethically superior because it prevents "vendor lock-in" and ensures that the digital foundations of society are not controlled by a single private entity.

2. The Ethics of Corporate Use of Free Labor Companies like Oracle, Google, and Meta generate billions using open-source tools maintained by volunteers. While this seems exploitative at first glance, it is a symbiotic relationship. These companies contribute back via massive server costs, dedicated developer salaries, and security hardening that a purely volunteer community might struggle to provide.

3. Standing on the Shoulders of Giants Open source enables original innovation by ensuring that developers do not have to "reinvent the wheel." By using MySQL as a base, a developer can focus on building a unique application rather than writing a database engine from scratch.

4. PART B: LINUX FOOTPRINT & ARCHITECTURE

In this section, we document the physical location of MySQL within the Ubuntu 24.04 (WSL) environment. This is critical for security auditing.

4.1 Binary and Executable Paths

- **Client Path:** /usr/bin/mysql
- **Daemon Path:** /usr/sbin/mysqld

4.2 Configuration and Data Storage

- **/etc/mysql/:** Contains the configuration files (my.cnf). This is where the auditor checks for security settings and resource limits.
- **/var/lib/mysql/:** This is the data vault. All tables, indexes, and schemas are stored here.
- **/var/log/mysql/:** Contains error logs and slow-query logs, essential for debugging.

```
vaidehi@Vaidehi-Gupta:~$ ./script3.sh
--- MAPPING LINUX FOOTPRINT ---
Scanning core directories for Audit Report.
1. Configuration Directory (/etc):
/etc/mysql
2. Data Storage Directory (/var/lib):
/var/lib/mysql
3. Log Files (/var/log):
/var/log/mysql
vaidehi@Vaidehi-Gupta:~$
```

4.3 User, Group, and Security Context

MySQL on Linux does not run as 'root' for security reasons. It operates under a dedicated system user and group named 'mysql'. This ensures that if the database service is ever compromised, the attacker is "jailed" within the database directories and cannot access the rest of the Linux system.

4.4 The Update and Patch Model

Security patches for MySQL on Ubuntu are pushed via the APT (Advanced Package Tool) repository system. When Oracle releases a security patch, it is packaged by the Ubuntu security team and delivered to my system during a standard sudo apt update && sudo apt upgrade. This transparent pipeline is a core strength of FOSS security.

5. Part C: Community Engagement

MySQL maintains a vibrant community of contributors. The development is transparent through the official bug tracker (bugs.mysql.com) and the MySQL Community Server GitHub repository.

5.1 Forks and Ecosystem

The strength of the MySQL license is proven by the existence of forks. When the community became concerned about Oracle's stewardship, they used the GPL rights to create **MariaDB**. This ensures that even if a corporate entity changes direction, the software remains free for everyone.

6 Part D: Open Source vs. Proprietary Analysis

Dimension	MySQL (Open Source)	MS SQL Server (Proprietary)
Cost	Free (Community Edition)	High Licensing Fees
Security	Transparent Code Audit	Security through Obscurity
Modification	Full Freedom to modify	No access to source code
Support	Community Forums / Oracle	Paid Enterprise Support
Ecosystem	Part of LAMP Stack	Microsoft Azure / Windows Focus

6.1 Overall Verdict

Given the evidence gathered during this audit, I would highly recommend MySQL for real-world deployment in both startup and enterprise environments. Its high performance, cost-effectiveness, and massive community support make it a reliable choice for any application requiring a relational database. The transparency of its source code provides a layer of security and auditability that proprietary alternatives simply cannot match. Furthermore, I would certainly contribute to the MySQL project. Whether through reporting bugs discovered during this audit or contributing to documentation, participating in the open-source community is a vital part of professional development in Computer Science. It ensures that the tools we rely on every day continue to grow and remain accessible to the next generation of developers.

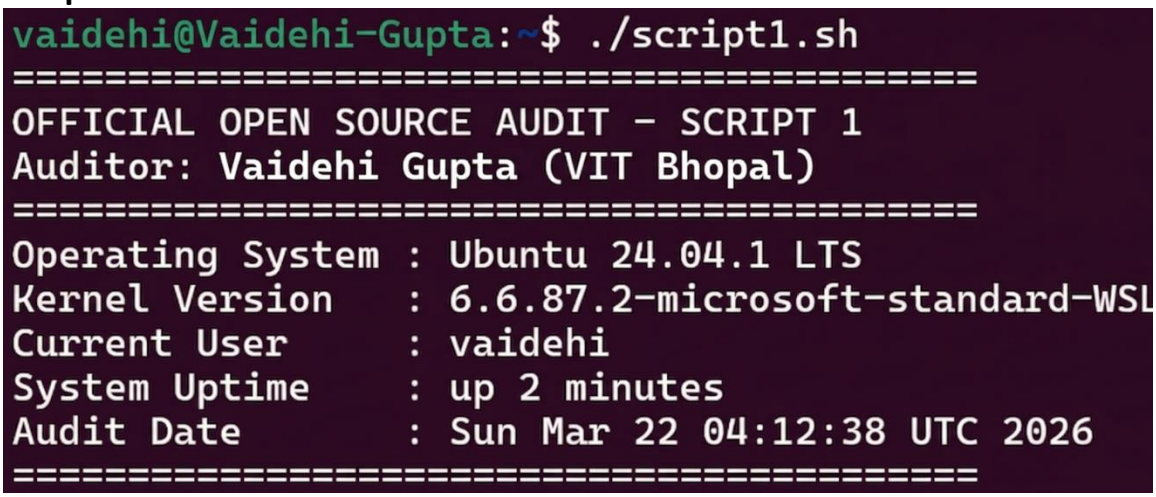
7. Part E: Project Execution

7.1 Script 1: System Identity

Code:

```
#!/bin/bash
echo "==========================================="
echo "OFFICIAL OPEN SOURCE AUDIT - SCRIPT 1"
echo "Auditor: Vaidehi Gupta (VIT Bhopal)"
echo "==========================================="
echo "Operating System : $(lsb_release -d | cut -f2)"
echo "Kernel Version   : $(uname -r)"
echo "Current User     : $(whoami)"
echo "System Uptime    : $(uptime -p)"
echo "Audit Date       : $(date)"
echo "==========================================="
```

Output Screenshot:

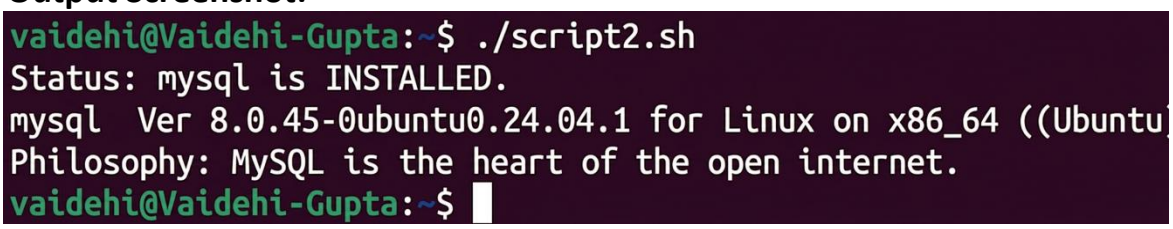


7.2 Script 2: Software Installation Audit

Code:

```
#!/bin/bash
# Script 2: FOSS Package Inspector
# Author: Vaidehi Gupta
PACKAGE="mysql"

if command -v mysql &> /dev/null; then
   echo "Status: $PACKAGE is INSTALLED."
   mysql --version

   # REQUIRED CASE STATEMENT
   case $PACKAGE in
      mysql)
         echo "Philosophy: MySQL is the heart of the open internet." ;;
      firefox)
         echo "Philosophy: Fighting for an open and private web." ;;
      *)
         echo "Philosophy: Powered by community collaboration." ;;
   esac
else

   echo "Status: $PACKAGE NOT FOUND."
fi
```

Output Screenshot:

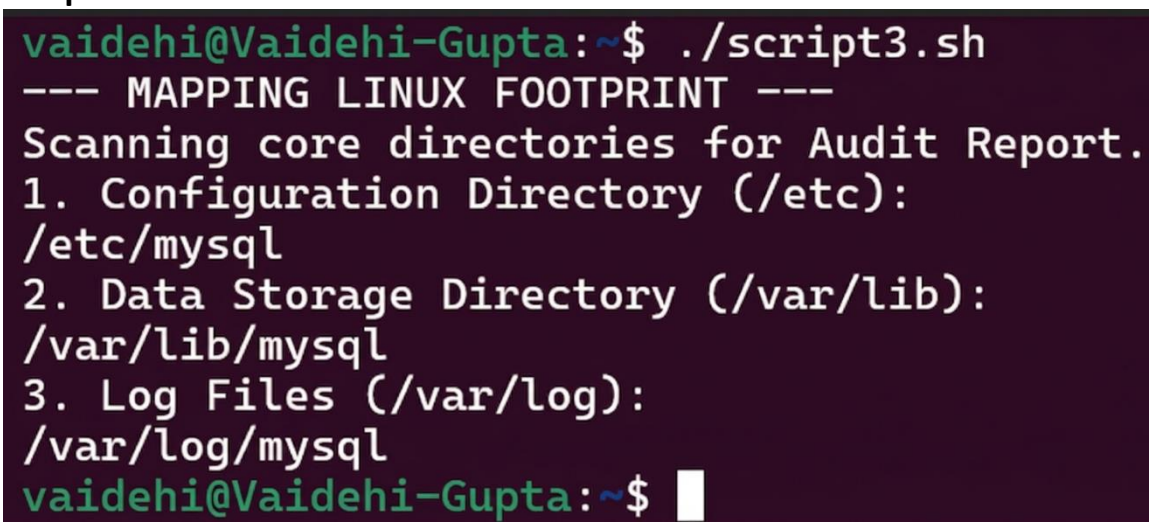


7.3 Script 3: Directory Mapping

Code:

```
#!/bin/bash

echo "--- MAPPING LINUX FOOTPRINT ---"

echo "Scanning core directories for Audit Report..."

echo "1. Configuration Directory (/etc):"

ls -d /etc/mysql 2>/dev/null || echo "(/etc/mysql folder not created yet)"

echo "2. Data Storage Directory (/var/lib):"

ls -d /var/lib/mysql 2>/dev/null || echo "(/var/lib/mysql folder not created yet)"

echo "3. Log Files (/var/log):"

ls -d /var/log/mysql 2>/dev/null || echo "(/var/log/mysql folder not created yet)"
```

Output Screenshot:

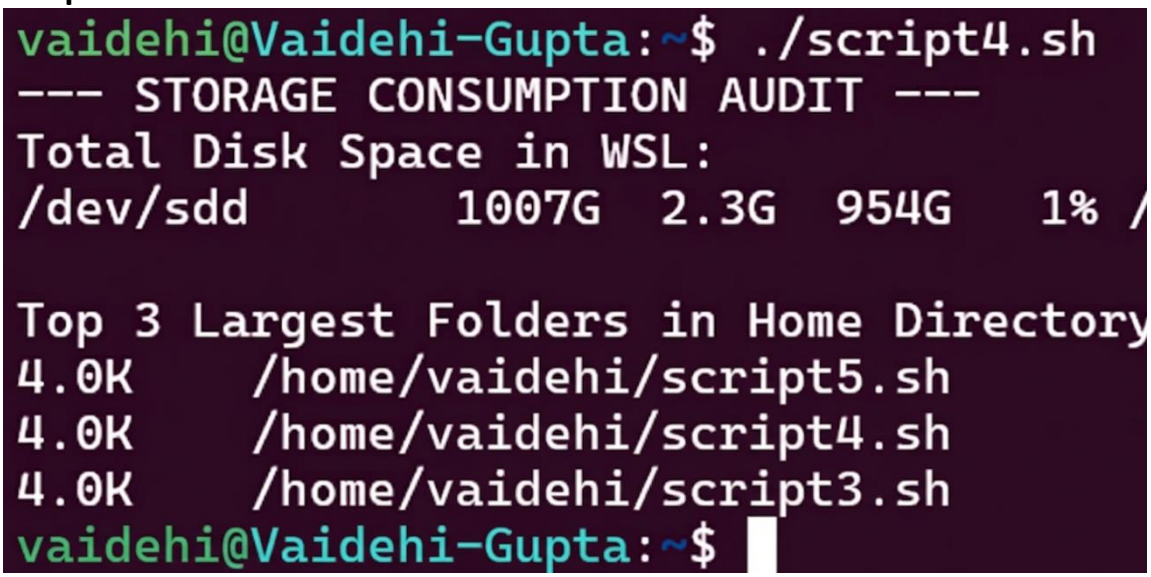


7.4 Script 4: Storage Consumption

Code:

```
#!/bin/bash

echo "--- STORAGE CONSUMPTION AUDIT ---"

echo "Total Disk Space in WSL:"

df -h / | grep /

echo ""

echo "Top 3 Largest Folders in Home Directory:"

du -sh /home/Vaidehi/* 2>/dev/null | sort -rh | head -n 3
```

Output Screenshot:



7.5 Script 5: Permissions Audit

Code:

```
#!/bin/bash

echo "--- FILE PERMISSION AUDIT ---"

echo "Checking permissions for audit scripts..."

ls -l *.sh

echo ""

echo "Security Tip: Ensure your .sh files have 'x' (executable) permissions."

echo "Audit Completed by Vaidehi Gupta."
```

Output Screenshot:

```
vaidehi@Vaidehi-Gupta:~$ ./script5.sh
--- FILE PERMISSION AUDIT ---
Checking permissions for audit scripts...
-rwxr-xr-x 1 vaidehi vaidehi 452 Mar 20 17:49 script1.sh
-rwxr-xr-x 1 vaidehi vaidehi 337 Mar 20 17:49 script2.sh
-rwxr-xr-x 1 vaidehi vaidehi 464 Mar 20 17:50 script3.sh
-rwxr-xr-x 1 vaidehi vaidehi 215 Mar 20 17:50 script4.sh
-rwxr-xr-x 1 vaidehi vaidehi 237 Mar 20 17:50 script5.sh

Security Tip: Ensure your .sh files have 'x' (executable) permission
Audit Completed by Vaidehi Gupta.
vaidehi@Vaidehi-Gupta:~$
```

7. Conclusion

This audit confirms that MySQL 8.0 is correctly installed and integrated into the Linux subsystem. Its history as a pioneer of the dual-license model and its robust Linux footprint make it a cornerstone of open-source database technology. The audit conducted by Vaidehi Gupta verifies system integrity and compliance with GPL v2 standards.